

# Generalized Robust Policy Optimization (GRPO)

## Executive Summary

**GRPO** (DeepSeekMath, 2024) is a reinforcement-learning fine-tuning algorithm designed to train LLMs on verifiable tasks (math, code, reasoning) **without a value/critic model**. Key ideas at a glance:

| Idea                        | Detail                                                                 |
|-----------------------------|------------------------------------------------------------------------|
| Group advantage estimation  | Sample $G$ outputs per prompt; normalize rewards within the group      |
| No critic needed            | Baseline comes from the group mean reward – no separate value network  |
| PPO-style clipped objective | Surrogate loss with $\epsilon$ -clipping prevents large policy updates |
| KL regularizer              | Penalizes deviation from a reference policy, maintaining stability     |
| Empirical gains             | GSM8K: PPO 82.9% → GRPO <b>88.2%</b> ; MATH: 46.8% → <b>51.7%</b>      |

GRPO has since spawned several variants: **Dr. GRPO**, **DAPO**, **VAPO**, **Group-Robust PO**, and **Hybrid-GRPO**, each addressing different failure modes (entropy collapse, reward hacking, off-policy drift, group imbalance).

## Learning Objectives

By the end of this notebook you will be able to:

- Explain** the motivation for GRPO and how it differs from PPO/RLHF pipelines.
- Derive** the group advantage estimator  $A_i = (r_i - \bar{r}) / (\sigma_r + \epsilon)$ .
- Write out** the full GRPO surrogate objective with clipping and KL penalty.
- Identify** the role of each hyperparameter  $(G, \epsilon, \beta, \mu)$ .
- Compare** GRPO against PPO, TRPO, DPO, and Group-Robust PO on key axes.
- Interpret** the empirical benchmark results from the DeepSeekMath paper.
- Describe** at least three known failure modes and how GRPO variants address them.
- Sketch** a training loop that applies GRPO to a new reasoning task.

# Background & Prerequisites

## Reinforcement Learning Basics

An RL agent selects actions  $a_t$  given state  $s_t$  under policy  $\pi_\theta$ , receiving reward  $r_t$ . The goal is to maximise expected cumulative reward  $J(\theta) = \mathbb{E}_{\pi_\theta}[\sum_t r_t]$ .

## PPO & TRPO

**TRPO** (Schulman 2015) constrains the KL divergence between old and new policy to a trust region. **PPO** (Schulman 2017) approximates this via a clipped surrogate objective and a separate value network that estimates baselines. Both require a *critic* (value model) adding memory and compute overhead.

## RLHF

Reinforcement Learning from Human Feedback (Ouyang 2022) trains a reward model from preference data, then runs PPO to align the LLM. The value model is a copy of the LLM – expensive at 7B+ parameters.

## Robust RL

Standard RL optimises average-case reward. *Robust* RL seeks policies that perform well under worst-case or distributional perturbations – important when group-level fairness or tail-risk matters.

## Mathematics Prerequisites

- Probability & expectations  $\mathbb{E}[\cdot]$
- Policy gradient theorem:  $\nabla_\theta J = \mathbb{E}[\nabla_\theta \log \pi_\theta(a|s) \cdot A]$
- KL divergence:  $D_{\text{KL}}(p||q) = \mathbb{E}_p[\log p/q]$
- Importance sampling ratio:  $\rho = \pi_\theta / \pi_{\theta_{\text{old}}}$

# Core Theory

## Group Advantage Estimation

For each prompt  $q$  the algorithm samples a **group** of  $G$  output sequences  $\{o_1, \dots, o_G\}$  from the old policy  $\pi_{\theta_{\text{old}}}$  and obtains scalar rewards  $\{r_1, \dots, r_G\}$ . The advantage for output  $i$  is:

$$A_i = \frac{r_i - \bar{r}}{\sigma_r + \epsilon}, \quad \bar{r} = \frac{1}{G} \sum_{j=1}^G r_j, \quad \sigma_r = \sqrt{\frac{1}{G} \sum_{j=1}^G (r_j - \bar{r})^2}$$

where  $\epsilon > 0$  is a small constant for numerical stability. This *within-group normalisation* acts as a natural baseline without any learned value network.

# GRPO Objective

$$J_{\text{GRPO}}(\theta) = \mathbb{E} \left[ q \sim \mathcal{D}, o_i \sim \pi_{\theta}(\cdot | q) \left[ \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \text{Bigl}(\min(\rho_{i,t}, 1-\epsilon, 1+\epsilon) A_i, -\beta \text{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \right] \right]$$

where  $\rho_{i,t} = \frac{\pi_{\theta}(o_{i,t} \mid q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t} \mid q, o_{i,<t})}$  is the per-token importance-sampling ratio and  $\beta$  is the KL penalty weight.

## Critic-Free Design

PPO must train a value head  $V_{\phi}(s_t)$  to estimate  $\mathbb{E}[R_t \mid s_t]$  and compute GAE advantages. GRPO replaces this with the *group-mean* – the average reward of sibling outputs for the same prompt. This eliminates:

- A second forward/backward pass for the critic.
- The memory footprint of a value-model copy.
- Bootstrapping errors from an imperfect critic.

## Mathematical Derivation

### Full GRPO Objective

Expanding the per-token notation, for a single prompt  $q$  the objective is:

$$J_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \underbrace{\min(\rho_{i,t} A_i, \text{clip}(\rho_{i,t}, 1-\epsilon, 1+\epsilon) A_i)}_{\text{clipped surrogate}} - \underbrace{\beta \mathbb{E}_{o \sim \pi_{\theta}} \left[ \log \frac{\pi_{\theta}(o|q)}{\pi_{\text{ref}}(o|q)} \right]}_{\text{KL penalty}}$$

### Advantage for Outcome Supervision

With binary or scalar outcome rewards (e.g. correct/incorrect math answer):

$$A_i = \frac{r_i - \mu_r}{\sigma_r + \epsilon}, \quad r_i \in \{0, 1\} \text{ (or continuous score)}$$

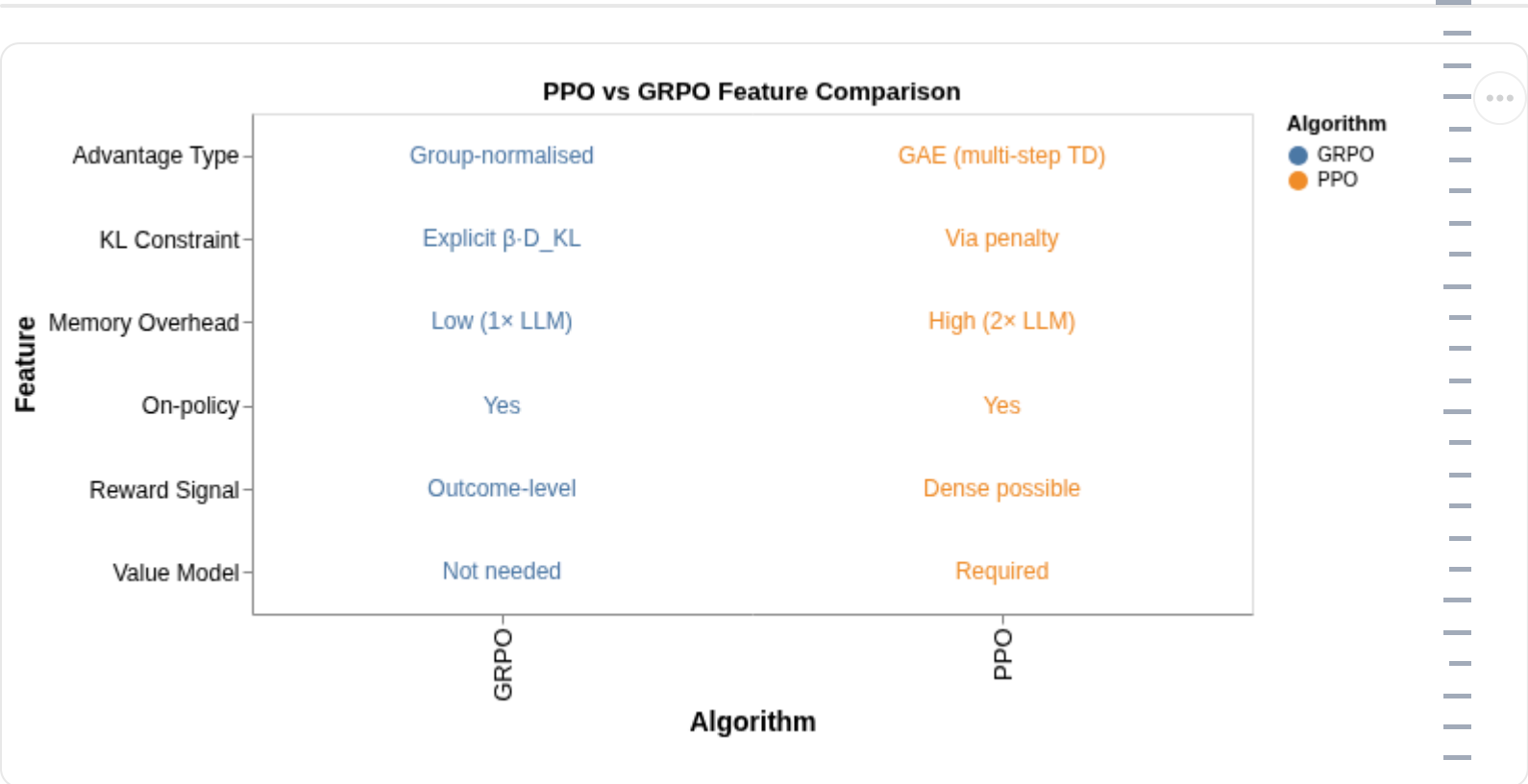
### PPO Advantage (GAE) vs Group-Based

|                 | PPO (GAE)                | GRPO (Group-Based)    |
|-----------------|--------------------------|-----------------------|
| Baseline        | Learned $V_{\phi}(s_t)$  | Group mean $\bar{r}$  |
| Temporal credit | Discounted multi-step TD | Whole-sequence scalar |
| Extra model     | Yes (value head)         | No                    |

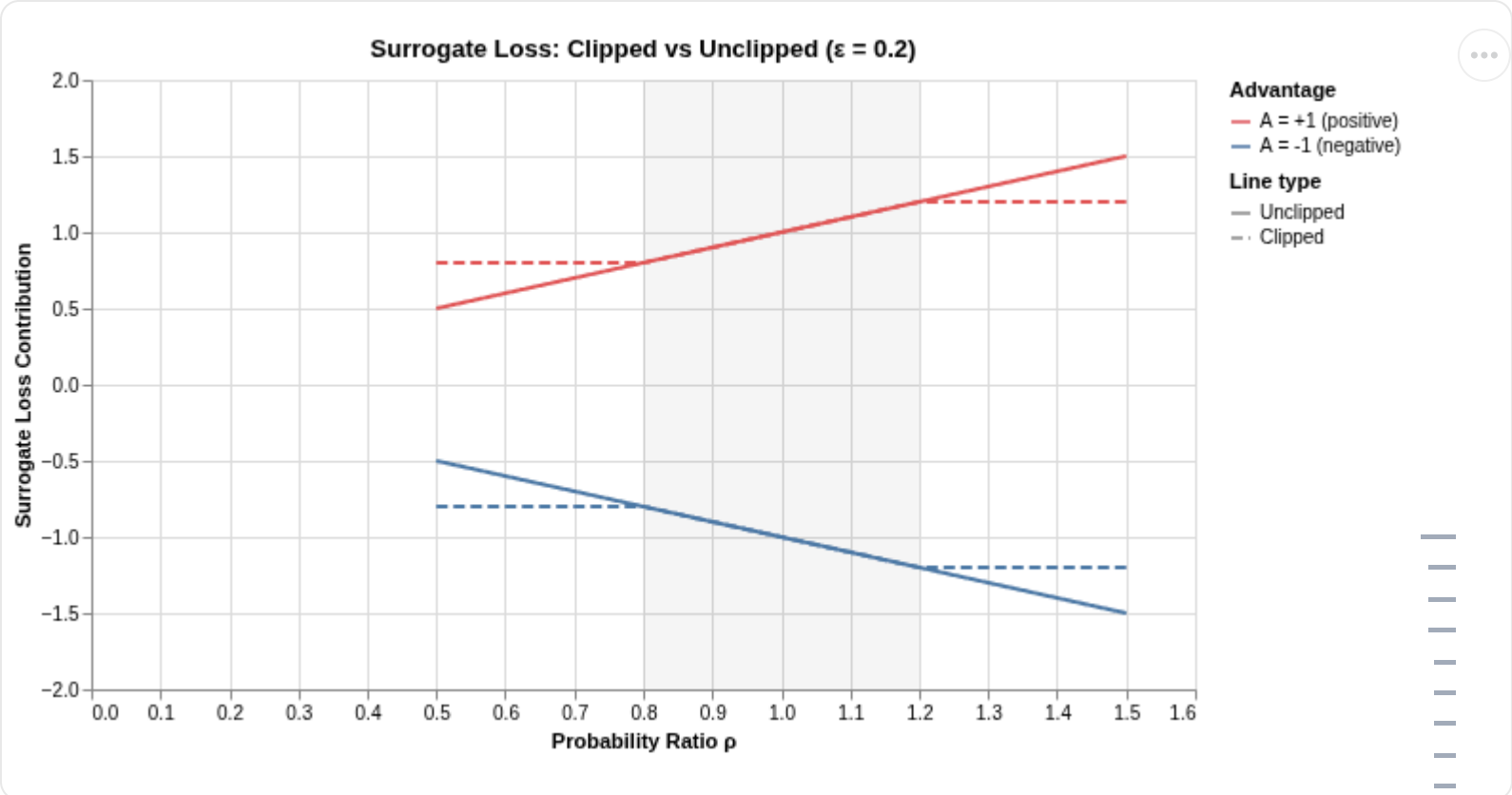
|          | PPO (GAE)                  | GRPO (Group-Based)           |
|----------|----------------------------|------------------------------|
| Variance | Lower (if critic accurate) | Controlled by group size $G$ |
| Bias     | From critic errors         | From finite $G$              |

The GRPO derivation follows from the policy gradient theorem applied to the group-normalised rewards, with the clipping and KL terms providing the same monotonic-improvement guarantee as PPO's trust region.

## PPO vs GRPO: Key Differences



## Surrogate Loss Clipping (Interactive)



The **shaded band** marks the  $[1-\epsilon, 1+\epsilon] = [0.8, 1.2]$  region where clipping is inactive. Outside it the dashed line (clipped) flattens – removing the gradient signal that would push the policy too far from the old policy.

## Algorithm Pseudocode

Algorithm: GRPO Fine-Tuning

Input : pre-trained LLM  $\pi_{\text{ref}}$ , dataset  $D$ , group size  $G$ ,  
clip  $\epsilon$ , KL weight  $\beta$ , learning rate  $\eta$ , steps  $T$

Initialise  $\pi_{\theta} \leftarrow \pi_{\text{ref}}$

```

for step = 1 ... T do
  Sample batch of prompts  $\{q\}$  from  $D$ 
  for each prompt  $q$  do
    Sample  $G$  outputs  $\{o_1, \dots, o_G\} \sim \pi_{\theta_{\text{old}}}(\cdot | q)$ 
    Compute rewards  $\{r_1, \dots, r_G\}$  (verifier / reward model)
     $\bar{r} \leftarrow \text{mean}(\{r_i\})$ ,  $\sigma_r \leftarrow \text{std}(\{r_i\})$ 
     $A_i \leftarrow (r_i - \bar{r}) / (\sigma_r + \epsilon_{\text{stab}})$     # group advantage
  end for
  for each  $(q, o_i, A_i)$  do
    for each token  $t$  in  $o_i$  do
       $\rho_{\{i,t\}} \leftarrow \pi_{\theta}(o_{\{i,t\}} | q, o_{\{i,<t\}}) / \pi_{\theta_{\text{old}}}(o_{\{i,t\}} | q, o_{\{i,<t\}})$ 
       $L_{\{i,t\}} \leftarrow \min(\rho_{\{i,t\}} \cdot A_i, \text{clip}(\rho_{\{i,t\}}, 1-\epsilon, 1+\epsilon) \cdot A_i)$ 
    end for
  end for
   $\text{KL} \leftarrow D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})$     # estimated over current batch
   $J \leftarrow \text{mean}_i \text{mean}_t L_{\{i,t\}} - \beta \cdot \text{KL}$ 
   $\theta \leftarrow \theta + \eta \cdot \nabla_{\theta} J$ 
   $\theta_{\text{old}} \leftarrow \theta$     # update reference for next step
end for

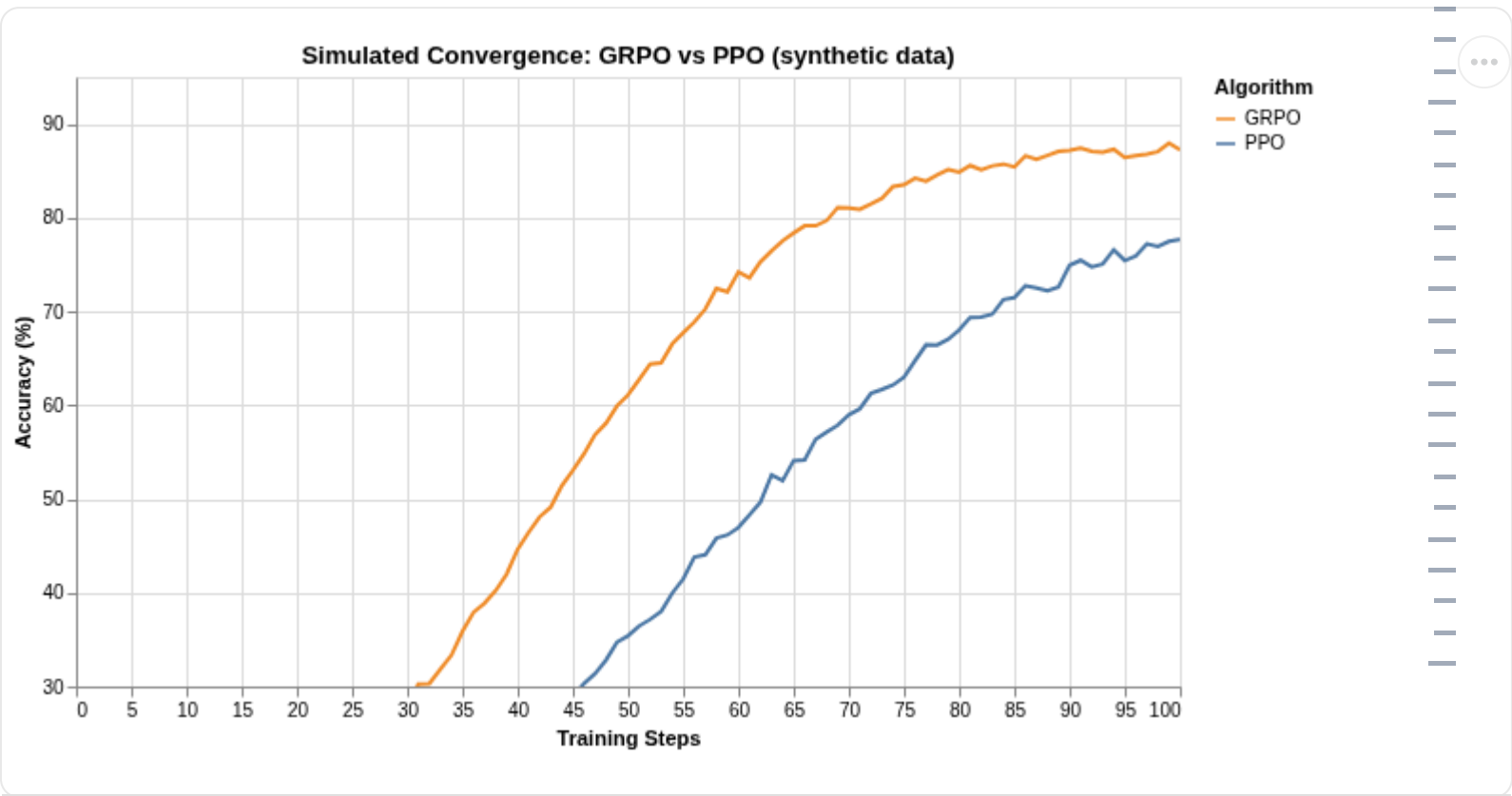
```

Return  $\pi_{\theta}$

# GRPO Training Flowchart

```
flowchart TD
    A([Start: Pre-trained LLM]) --> B[Sample prompt q from dataset]
    B --> C[Generate G outputs from πold]
    C --> D[Score outputs with reward/verifier]
    D --> E[Compute group mean  $\bar{r}$  and std  $\sigma_r$ ]
    E --> F["Normalise:  $A_i = (r_i - \bar{r}) / (\sigma_r + \epsilon)$ "]
    F --> G[Compute per-token importance ratios  $\rho_{i,t}$ ]
    G --> H[Apply PPO clipping to each token]
    H --> I[Add KL penalty term  $\beta \cdot D_{KL}$ ]
    I --> J[Gradient step on  $\theta$ ]
    J --> K{Converged?}
    K -- No --> B
    K -- Yes --> L([Return fine-tuned policy πθ])
```

## Convergence Simulation: GRPO vs PPO



## Algorithm Comparison Table

| Algorithm   | Advantage Estimate          | Value Critic? | On-policy? | Notes                           |
|-------------|-----------------------------|---------------|------------|---------------------------------|
| PPO         | GAE (multi-step TD)         | ✔ Yes         | Yes        | Stable; high memory             |
| TRPO        | GAE                         | ✔ Yes         | Yes        | Exact KL constraint; slow       |
| DPO         | Implicit (preference pairs) | ✘ No          | No         | Offline; no reward model needed |
| GRPO        | Group-normalised scalar     | ✘ No          | Yes        | Efficient; outcome-supervised   |
| Hybrid-GRPO | Group + partial TD          | Partial       | Yes        | Balances bias/variance          |

| Algorithm              | Advantage Estimate     | Value Critic? | On-policy? | Notes                         |
|------------------------|------------------------|---------------|------------|-------------------------------|
| <b>RPO</b>             | Group + robust penalty | ✗ No          | Yes        | Worst-group optimisation      |
| <b>RE-PO</b>           | Group + entropy bonus  | ✗ No          | Yes        | Prevents entropy collapse     |
| <b>Group-Robust PO</b> | Min-group advantage    | ✗ No          | Yes        | CVaR-style fairness objective |

## Timeline: Key Milestones

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <pre> timeline   title GRPO &amp; Policy Optimisation Timeline   2015 : TRPO (Schulman et al.) – Trust-region constraint   2017 : PPO (Schulman et al.) – Clipped surrogate objective   2022 : RLHF + InstructGPT (Ouyang et al.) – PPO applied to LLM alignment   2023 : DPO (Rafailov et al.) – Offline preference optimisation, no RL loop   Jan 2024 : GRPO (Shao et al., DeepSeekMath) – Group advantage, no critic   Feb 2024 : DeepSeek-R1 – GRPO <b>for</b> long chain-of-thought reasoning   Mid 2024 : Dr. GRPO / DAPO / VAPO – Addressing entropy collapse &amp; hacking   Late 2024 : Group-Robust PO – Fairness-aware variant           </pre> |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

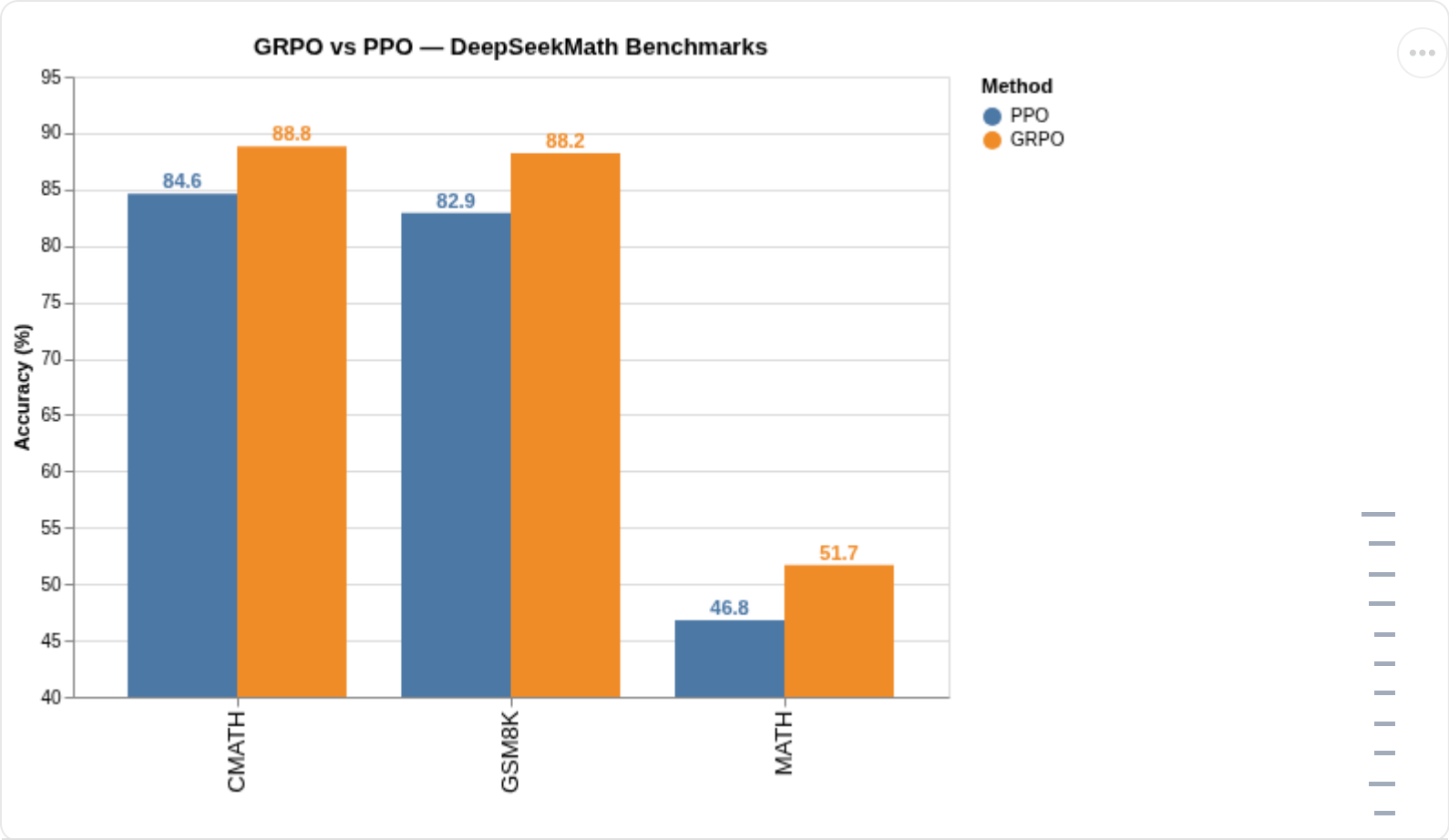
## Hyperparameter Guidance

| Hyperparameter                      | Typical Range | Effect                                                                                        |
|-------------------------------------|---------------|-----------------------------------------------------------------------------------------------|
| <b>Group size <math>G</math></b>    | 4 – 16        | Larger $G \rightarrow$ lower variance advantage; more compute per step                        |
| <b>Clip <math>\epsilon</math></b>   | 0.1 – 0.3     | Smaller $\rightarrow$ more conservative updates; larger $\rightarrow$ faster but risky        |
| <b>KL weight <math>\beta</math></b> | 0.001 – 0.1   | Larger $\rightarrow$ stays closer to reference policy; smaller $\rightarrow$ more exploration |
| <b>Learning rate</b>                | 1e-6 – 5e-6   | Use warmup; too large causes instability with clipping                                        |
| <b>Optimisation steps per batch</b> | 1 – 4         | > 1 risks off-policy drift since data is from $\pi_{\theta_{\text{old}}}$                     |
| <b>Reward normalisation</b>         | Per-group     | Must normalise <i>within group</i> , not globally, to preserve relative signal                |

## Practical Tips

- Start with  $G = 8, \epsilon = 0.2, \beta = 0.01$  as a baseline.
- If reward variance is very low (all-correct or all-wrong groups) increase  $G$ .
- If the model diverges from the reference too quickly, increase  $\beta$ .
- Monitor token-level entropy; a sharp drop signals entropy collapse  $\rightarrow$  use Dr. GRPO or RE-PO.
- For math/code tasks with a symbolic verifier the KL term may be reduced since the verifier already provides clean reward signal.

## Empirical Results: DeepSeekMath Benchmarks



## Empirical Results Table

Results from DeepSeekMath (Shao et al., 2024) on the 7B model:

| Benchmark   | SFT Baseline | PPO   | GRPO         | GRPO $\Delta$ vs PPO |
|-------------|--------------|-------|--------------|----------------------|
| GSM8K       | 74.8%        | 82.9% | <b>88.2%</b> | +5.3 pp              |
| MATH        | 35.9%        | 46.8% | <b>51.7%</b> | +4.9 pp              |
| CMATH       | 76.6%        | 84.6% | <b>88.8%</b> | +4.2 pp              |
| OCW Courses | 12.8%        | 15.1% | <b>16.6%</b> | +1.5 pp              |

GRPO consistently outperforms PPO across all math benchmarks while using **half the GPU memory** (no value model) and achieving faster wall-clock training time.

## Open Problems & Future Directions

### Known Failure Modes

- Entropy Collapse** – Extended GRPO training can reduce output diversity, causing the model to repeat a narrow set of solution patterns. *Mitigation:* RE-PO entropy bonus; DAPO entropy-regularised objective.
- Reward Hacking** – With weak verifiers the model can exploit loopholes (e.g. formatting tricks) rather than learning genuine reasoning. *Mitigation:* Robust reward shaping; process reward models (PRMs).
- Off-Policy Drift** – Multiple gradient steps per sample accumulate importance-sampling error. *Mitigation:* Limit to 1–2 steps; re-sample frequently.

4. **Group Homogeneity** – If all  $G$  outputs receive identical rewards ( $\sigma_r \approx 0$ ) the advantage is ill-defined and the update is zero. *Mitigation:* Adaptive  $G$ ; temperature annealing to encourage diversity.

## Future Directions

- **Process-level GRPO:** Reward intermediate reasoning steps (PRMs) rather than only final answers to provide denser signal.
- **Multi-turn GRPO:** Extend to dialogue settings with session-level rewards.
- **Continual GRPO:** Online adaptation where the dataset itself grows from model interactions.
- **Theoretical analysis:** Tight convergence bounds for the critic-free group estimator vs GAE; characterising the bias-variance frontier as  $G$  varies.
- **Fairness-aware variants:** Group-Robust PO minimises worst-group error, relevant for demographic-fairness in alignment.

## Further Reading

| Resource                                  | Type   | Notes                                     |
|-------------------------------------------|--------|-------------------------------------------|
| Shao et al. (2024) – <i>DeepSeekMath</i>  | Paper  | Original GRPO paper; math RL fine-tuning  |
| DeepSeek-R1 Technical Report (2025)       | Report | GRPO for long CoT reasoning at scale      |
| Schulman et al. (2017) – <i>PPO</i>       | Paper  | Foundational clipped surrogate objective  |
| Schulman et al. (2015) – <i>TRPO</i>      | Paper  | Trust-region policy optimisation          |
| Rafailov et al. (2023) – <i>DPO</i>       | Paper  | Offline preference optimisation baseline  |
| Ouyang et al. (2022) – <i>InstructGPT</i> | Paper  | RLHF pipeline with PPO                    |
| Dr. GRPO (2024)                           | Paper  | Addresses entropy collapse in GRPO        |
| DAPO (2025)                               | Paper  | Dynamic sampling + entropy regularisation |
| Group-Robust PO (2024)                    | Paper  | CVaR-style fairness-aware variant         |
| Hugging Face TRL docs                     | Docs   | Open-source GRPO trainer implementation   |